

LESSON 5-8 • SESSION 2 • PRACTICE • FORM B**Lateral Area of Pyramids**

Learning target – I can find the lateral area of a pyramid, which covers only its slanted faces. • 6.G.A.4

REMEMBER

- **Lateral area** = the slanted faces only. Do **not** include the base.
- **Surface area** = lateral area + base area.
- A triangle-base pyramid has **3** slanted faces; a square pyramid has **4**.

WORKED EXAMPLE

Square base 4 m; slant height 6 m; only the sides are covered.

Step 1 Leave out the base.**Step 2** One triangle: $\frac{1}{2} \times 4 \times 6 = 12 \text{ m}^2$.**Step 3** Four triangles: $4 \times 12 = 48$.**Answer lateral area = 48 m²****WORDS** lateral area (*área lateral*) • surface area (*área superficial*) • slant height (*altura inclinada*)**PRACTICE**

show your work • label units

1. Find each lateral area.

a. 4 faces, each 30 cm^2 ____ cm^2

b. SA = 180 in^2 , base = 64 in^2 ____ in^2

c. square base 5 m, slant 8 m ____ m^2

d. square base 12 in, slant 9 in ____ in^2

- 2.
- Canvas Frame.**
- Jamal's frame is a pyramid with a triangle base (18 in sides) and slant height 10 in. Only the slanted faces get canvas.

a. Faces that get canvas _____

b. One slanted face ____ in^2

c. Canvas needed ____ in^2

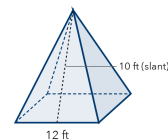
Show your work.

- 3.
- Open-Bottom Tent.**
- This tent is a square pyramid. The floor is open ground.

a. One panel ____ ft^2

b. Fabric for the tent ____ ft^2

c. Total if the floor were covered ____ ft^2



Show your work.

- 4.
- Gazebo Roof.**
- A square pyramid roof has base 16 ft and slant height 9 ft. Only its slanted faces get shingles.

Show your work.

Answer _____ ft^2

- 5.
- Tower Cap.**
- A square pyramid cap has base 18 m and slant height 15 m. Only its slanted faces are copper.

Show your work.

Answer _____ m^2

- 6. Display Pyramid.** A display is a square pyramid: base 20 in, slant height 24 in. Every face is wrapped, base included. How much paper?

Hint Include the base.

Show your work.

Answer _____ in²

- 7. Luis's Canvas.** A square pyramid has base 24 cm and slant height 15 cm. Luis says the four slanted faces cover 1,440 cm². Is Luis right?

Hint Find one slanted face first.

Show your work.

Answer _____ cm²

- 8. Tent Cover.** A shelter is a square pyramid: base 12 ft, slant height 8 ft, open floor. Canvas costs \$4 per ft².

Hint Find the lateral area first, then multiply by the price.

- a. Lateral area _____ ft²
 b. Cost of the canvas _____
 c. Total SA; why use lateral area here? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Grace's Work.** A square pyramid has base 12 cm and slant height 7 cm. Only the slanted faces are painted. Grace wrote: "Area = 144 + 168 = 312 cm²." Explain the mistake and fix it.

TEST PRACTICE

choose the best answer

- 10.** Total surface area equals lateral area plus the –
A base area **B** slant height **C** apex **D** one triangle
- 11.** How many slanted faces does a square pyramid have?
A 3 **B** 4 **C** 5 **D** 6

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

1,440 Q	192 G	720 A	288 V	768 E	540 E	240 O	864 Z	1,360 R	270 C
#2c	#3b	#4	#5	#6	#7	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____